> formula using either distances along magnetic north and east or distances along true north and east, so also between any two *events* in spacetime there is an *absolute straight-line interval* computable from an analogue of Pythagoras's formula using lengths and times measured in either reference frame, mine or yours.

It was this analogue of Pythagoras's formula (I shall call it *Minkowski's formula)* that led Hermann Minkowski to his discovery of absolute spacetime.

The details of Minkowski's formula will *not* be important in the rest of this book. There is no need to master them (though for readers who are curious, they are spelled out in Box 2.1). The only important thing is that events in spacetime are analogous to points in space, and there is an absolute interval between any two events in spacetime completely analogous to the straight-line distance between any two points on a flat sheet of paper. The absoluteness of this interval (the fact that its value is the same, regardless of whose reference frame is used to compute it) demonstrates that spacetime has an absolute reality; it is a four-dimensional fabric with properties that are independent of one's motion.

Box 2.1

Minkowski's Formula

You zoom past me in a powerful, 1-kilometer-long sports car, at a speed of 162,000 kilometers per second (54 percent of the speed of light); recall Figure 1.3. Your car's motion is shown in the following spacetime diagrams. Diagram (a) is drawn from your viewpoint; (b) from mine. As you pass me, your car backfires, ejecting a puff of smoke from its tailpipe; this backfire event is labeled *B* in the diagrams. Two microseconds (two-millionths of a second) later, as seen by you, a firecracker on your front bumper detonates; this detonation event is labeled *D.*